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Review

Digital Twin and Integrated Sensing and Communication Applications in Internet of Things: Enabling Communication Technologies, Taxonomy, Implications, and Open Issues

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Abstract

Future internet of things (IoT) services necessitate the integration of sensing and communication functions within the same system, utilizing digital twin (DT) technology. Integrated sensing and communication (ISAC) can address the need for widespread communication and high-precision sensing by leveraging the benefits of spectrum and hardware resource sharing. The DT represents a promising technology for achieving low latency and high energy efficiency, leveraging real-time monitoring, optimization, and predictive maintenance capabilities. Recent advancements in DT and ISAC for IoT systems research necessitate the establishment of effective communication technology methods between the physical entity and its digital representation. Most prior research reviews have not sufficiently explored the critical enabling technologies for DT and ISAC applications in IoT systems that facilitate communication between a physical entity and its digital counterparts. Previous research has primarily focused on the application of DT technology in IoT systems; however, little emphasis has been placed on the integration of ISAC technology with DT in IoT systems, as well as the important integrated sensing and communication technologies between physical entities and their digital counterparts. This paper presents a systematic literature review that focuses on the analysis of key communication technologies that facilitate connections in DT and ISAC for IoT systems. The implications of communication methods derived from the study are presented, focusing on technologies such as unmanned aerial vehicles (UAVs), reconfigurable intelligent surfaces (RISs), millimeter wave (mmWave), and massive MIMO (mMIMO) technologies. The emphasis on these technologies is due to their significance as essential enablers of integrated sensing and communication between physical entities and their digital counterparts. Furthermore, these technologies are critical for integrating DT and ISAC into the IoT systems to efficiently meet the needs of IoT services. Future discussions are finally addressing open research and the challenges that demand attention.

Keywords: digital twin; communication technologies; internet of things;integrated sensing and communication

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1. Introduction

The new IoT services, like haptics, brain-computer interaction (BCI), unmanned aerial vehicles (UAV), and the industrial metaverse, help bring together the digital and physical worlds [1–4]. These IoT services address a wide range of needs, including data

transfer, sensing, prediction, control, and decision-making. To address these different needs, it is necessary to equip DT and ISAC for IoT systems with cutting-edge technologies via two major trends: Digital Twin (DT) and Integrated Sensing and Communication (ISAC) [5–7]. IoT devices typically rely on connections that are intelligent, smooth, and pervasive to satisfy the various requirements of end-users.

ISAC is a design approach that integrates sensing and communication functions to enhance mutual benefits through the sharing of frameworks, spectrum resources, and signal processing [5,8–11]. However, IoT services have been rapidly changing demands regarding latency, data rates, energy efficiency (EE), dependability, and user-defined metrics, among others. To address these rapidly evolving requirements, it is essential to facilitate effective interactions among diverse IoT devices within IoT systems and technologies, which include wireless channel resources, security technologies, and computing technologies. As a result, constructing DT and ISAC for IoT systems that utilize IoT technologies is essential.

Designing DT and ISAC for IoT systems involves creating a digital twin that relies on the ISAC based on the IoT system [2]. A DT employs a virtual representation of the physical system that can be used for IoT applications [3,6,9,12]. A DT utilizes real-world inputs to generate predictions and make control decisions. In this context, alongside the digital representation of a physical system, a DT and ISAC for IoT systems will incorporate tools from sensing, communication, optimization theory, game theory, metaheuristics, and machine learning. Furthermore, to facilitate the transparent and irreversible management of data, a system based on DT may use blockchain technology [2]. A DT differs from conventional simulation tools in that it allows for examination of systems, processes, the impacts of various system variables through computer-based models and mathematical principles, and the real-time communication synchronizing the current states of the physical system and the DT.

The Importance of Integrating DT and ISAC in the IoT Development

ISAC is an essential enabler of IoT development. The sensing capability of ISAC can enhance security by determining the direction of prospective assailants, while the information sharing among various nodes facilitates a collaborative improvement in sensing performance [5]. Conversely, IoT systems utilize ISAC to enhance spectral efficiency, minimize interference, improve sensing and communication performance through collaborative technologies, and mitigate spectrum congestion [9,10].

Another major enabler for IoT development is DT, which collects data from IoT devices in real time to maintain representations [1]. Furthermore, DT provides real-time monitoring, simulation, and optimization of system processes, resulting in significant increases in system performance, cost-effectiveness, and predictive maintenance [12]. However, a key challenge in deploying DT for IoT systems is creating a reliable and secure communication channel between the physical and digital entities. ISAC represents a viable method to enable safe sensing and communication between the physical thing and its digital counterpart within the DT application in IoT systems.

The integration of DT and ISAC in IoT systems has the potential to fundamentally change industrial, commercial, and economic functions while simultaneously augmenting sensing and communication efficacy and enhancing the accuracy, dependability, and efficiency of real-time sensing data rather than addressing them in isolation. This integration facilitates reciprocal advantages for both DT and ISAC together with the IoT system.

Past studies have explored various applications of DT technology in the IoT systems. This article specifically focuses on the integration of ISAC with DT in IoT systems, emphasizing how these technologies facilitate the connection between physical entities and their digital counterparts [7]. While existing research primarily concentrated on DT

applications in IoT systems, there is a noticeable gap in academic studies that investigate ISAC technology's role in connecting these physical and digital entities. The overall goal of this study is to investigate the essential communication technologies that permit communication between the physical entity and its digital entity for efficient integration of DT and ISAC in IoT systems.

This systematic literature review intends to thoroughly analyze communication technologies that utilize DT and ISAC for IoT systems, with a particular focus on ISAC-based and DT-based schemes. We evaluated the evolution of publication types related to DT and ISAC for IoT systems, examined the types of publications that researchers have used to explore this topic, and identified which digital databases have the most papers on it. Additionally, we assessed the technologies that enable physical-to-virtual communication. We identified the following unique research question that would guide us in selecting relevant work.

(1) How have publication types connected to DT and ISAC for IoT systems evolved over the years?
(2) What types of publications have researchers used to investigate DT and ISAC for IoT systems?
(3) Which digital databases contain the most articles on this topic?
(4) What are the essential communication technologies that enable physical-to-virtual connections in DT and ISAC for IoT systems?

The primary contributions of our study are as follows:

1. Introducing a new taxonomy of DT and ISAC for IoT systems.
2. Discussing critical communications methods for DT and ISAC applications in IoT systems.
3. Exploration of common performance indicators used in DT and ISAC for the IoT field.
4. Investigating the impact of DT and ISAC applications on IoT development.

The subsequent sections of this paper are structured as follows: Section 2 provides an overview of the foundational concepts. Section 3 outlines the main differences of our paper to existing relevant survey papers. The methodology employed in this paper is detailed in Section 4. Section 5 outlines the primary findings from the systematic review along with a discussion of those results. Section 6 addresses several areas for future investigation. In conclusion, Section 7 sums up the findings presented in the paper.

2. Key Components of the Concerned DT and ISAC for IoT Systems

This section outlines the framework and features of a digital twin technology, alongside the system models of the ISAC, as well as an overview of the Internet of Things.

2.1. Internet of Things

The IoT anticipates a future when things are effortlessly connected to the internet, uniquely identifiable and accessible on the network, with their position and status readily available [13,14]. The IoT integrates the digital and physical domains into a cohesive platform enhanced with services and intelligence [14,15]. The extensive integration of the internet has become indispensable in daily life, and the IoT is emerging as a groundbreaking digital innovation. Regarding the forthcoming industrial revolution, IoT will enhance and optimize resource use, reduce human efforts, and conserve time [15]. Despite several benefits, IoT poses challenges such as interconnectivity, spectrum congestion, power consumption, high communication latency, and security [14–16]. As a result, in this study, we proposed integrating DT and ISAC technologies into IoT systems to overcome these difficulties [1,12,17].

2.2. Integrated Sensing and Communication: System Model and Performances Metrics

ISAC has recently garnered a lot of attention from industry and academia. As an illustration, the research presented in [5] examined the factors that were driving the formation of ISAC and presented a definition of the organization. In addition, the authors of [17] conducted additional research to investigate the state of the art in terms of cooperative communication and sensing. A full assessment of the history of ISAC can be found in [18], which is another point of interest. In a multi-user communication scheme, the authors of [19] proposed a beamforming method for joint radar sensing and communication. The researchers in [20] developed a detailed framework for the theoretical analysis of ISAC system performance. The progress in mobile communication technologies, including heterogeneous network (HetNet), millimeter wave (mmWave), terahertz (THz), massive MIMO, cell-free massive MIMO (CF-mMIMO), reconfigurable intelligent surface (RIS), stacked intelligent metasurfaces (SIM), and advanced signal processing algorithms, has facilitated the realization of ISAC [21–26]. Effective integration of ISAC with DT in the IoT systems can significantly enhance system performance and dependability by improving the quality of the channel. Additionally, performance metrics play a crucial role in measuring and evaluating the design, implementation, and optimization of this integration within the IoT context. Consequently, this section will review the channel modelling and performance metrics of the ISAC system as it relates to its integration with DT in IoT systems.

2.2.1. System Model

Channel modelling provides a basis for comprehending and forecasting the behavior of wireless signals as they propagate [22]. The establishment of a foundation for the management of sensor and communication resources is made possible by accurate channel modelling, which makes it easier to recognize and comprehend both beneficial and interference signals inside ISAC systems [22,23]. Furthermore, efficient interference control has the potential to improve the performance and dependability of the system by improving the quality of channel modelling in ISAC systems [24,26]. The channel modelling of ISAC can be categorized into two distinct types [27]: communication channel modelling and sensing channel modelling. The aim of communication channel modelling is to improve the precision of signal transmission, thereby neglecting scattering in intermediary processes. The sensing process depends on the reception of echo signals that are reflected from the target, which facilitates detection and parameter estimation [10]. Nonetheless, sensing and communication signal transmission adhere to the principles of electromagnetic propagation, thereby enabling a standardized approach to channel modelling. In [28], a stochastic ISAC channel model based on geometry that can be used with both mono-and bi-static sensing modes in the 3GPP channel mode scheme was suggested. Furthermore, the authors of [29] present a channel model for a near-field ISAC system that is more accurate. This goal is accomplished by considering the effects of effective aperture and polarization loss.

Based on the previous study, we define the modelling of communication and sensing channels, as divided into downlink and uplink sensing, as shown in Figure 1. An ISAC system usually includes a base station (BS), user equipment (UE), and targets that need to be sensed. The ISAC BS has a 2-D uniform planar array (UPA) placed in the xy plane. The dimensions of UPA is $S \times T$, characterized by uniform element intervals represented as d_a. The index (s, t) denotes the antenna element located in the sth row and the tth column. The phase difference between the (s, t)th element and the reference element, specifically, the $(0, 0)$th antenna element, is expressed as [9]

$$[\mathbf{a}(\theta, \phi)]_{s,t} = e^{\left[-j\frac{2\pi}{\lambda}d_a(s\cos\theta\sin\phi + t\sin\theta\sin\phi)\right]} \quad (1)$$

where θ and ϕ represent the azimuth and elevation angles, respectively, while λ denotes the wavelength. The steering matrix $\mathbf{A}(\theta, \phi)$ is the product of the steering vectors of the transmit array and receive array as follows [9]

$$[\mathbf{A}(\theta, \phi)]_{s,t} = \mathbf{a}_{Rx}(\theta, \phi)\mathbf{a}_{Tx}^{H}(\theta, \phi) \tag{2}$$

where $\mathbf{a}_{Rx}$ and $\mathbf{a}_{Tx}^{H}$ represent the steering vectors for the transmit and receive arrays, respectively, corresponding to the angle of arrival (AoA) for reception and the angle of departure (AoD) for transmission $\mathbf{a}_{Rx}$ and $\mathbf{a}_{Tx}$ are presented in the manner of $\mathbf{a}$ in Equation (1).

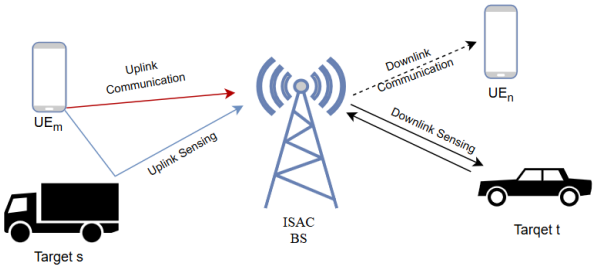


Figure 1. ISAC channel model.

(i) Communication Channel Modelling

The communication channel is defined as a unidirectional way of connecting the BS to the UE in both uplink and downlink modes. Consider that there exists a total of L paths, denoted by the index $l \in \mathcal{L} \triangleq \{0, 1, \ldots, L-1\}$, which includes one line-of-sight (LOS) path $(l = 0)$ and $(L-1)$ non-LOS (NLOS) paths. Due to the unidirectional nature of communication, the uplink and downlink transmissions are reciprocal. Thus the channel model for the kth user is represented as follows.

$$\begin{aligned}
&\mathbf{H}_k^{com}(t, \tau) \\
&= \sqrt{\frac{K}{K+1}}\mathbf{G}_0 e^{j2\pi f_{D_0}(t)t} e^{j2\pi f_c \tau_0} \mathbf{A}_{com}(\theta_0, \phi_0)\delta(\tau - \tau_0) \\
&+ \sum_{l=1}^{L-1} \sqrt{\frac{1}{K+1}}\mathbf{G}_l e^{j2\pi f_{D_l}(t)t} e^{j2\pi f_c \tau_l} \mathbf{A}_{com}(\theta_l, \phi_l)\delta(\tau - \tau_l)
\end{aligned} \tag{3}$$

where $\mathbf{G}_l$ is the large scale fading, K is the Rician factor, $f_{D_0}(t) = \left[\frac{\nu_0(t)}{\lambda}\right]$ and $\tau_0 = \left(\frac{D_0}{c}\right)$ are the Doppler frequency shift and delay of the LOS path, respectively. $f_{D_l}(t) = \left[\frac{\nu_l(t)}{\lambda}\right]$ and $\tau_l = \left(\frac{D_l}{c}\right)$ are the Doppler frequency shift and delay of the lth NLOS path, respectively. D_0, D_l, ν_0 and ν_l are the distances and radial velocities of the LOS path and the lth NLOS path, respectively. $\mathbf{A}_{com}$ represents the product of the steering vectors for the transmit and receive arrays, utilized for environmental target sensing, as shown in Equation (2).

(ii) Sensing Channel Modelling

Sensing channel modelling is divided into two types based on the modes of signal transmission and reception. In downlink sensing mode, the transmitter and receiver are deployed at the ISAC BS, creating a two-way channel. On the other hand, when in uplink sensing mode, the UE sends the ISAC signal to the BS, which is picked up by the BS after reflections from the targets. Here, communication channel and uplink sensing channel modelling are compatible. Consider that non-desired $P-1$ targets are spread across the environment. Subsequently, the targets may be indexed by $p \in \mathcal{P} \triangleq \{0, 1, \ldots, P\}$, with

$P = 0$ denoting the intended sensing target. The channel impulse responses (CIR) of downlink sensing can be written [9] as

$$\mathbf{H}^{sen}(t,\tau) = \sum_{p=0}^{P} \mathbf{G}_p e^{j2\pi f_{D_p}(t)t} e^{j2\pi f_c \tau_p} \mathbf{A}_{sen}(\theta_p, \phi_p)\delta(\tau - \tau_p) \quad (4)$$

where $\mathbf{G}_p = \left[\frac{\lambda^2 \sigma_{RCS,p}}{(64\pi^3 (D_p)^4)}\right]$ is the fading coefficient related to the pth target. $f_{D_p}(t) = \left[\frac{2v_p(t)}{\lambda}\right]$ and $\tau_p = \left(\frac{2D_p}{c}\right)$ are the Doppler frequency shift and delay of the pth target, respectively. $\sigma_{RCS,p}$, D_p, and v_p are radar cross section (RCS), distances and radial velocitiy of the pth target, respectively. $\mathbf{A}_{sen}(\theta_p, \phi_p)$ represents the product of the steering vectors for the transmit and receive arrays of ISAC BS in target and environmental cluster sensing.

The incorporation of ISAC with DT in IoT systems would enhance the channel quality between the physical entity and its digital counterparts. Consequently, ISAC channel modeling and performance metric indicators are essential for the integration of ISAC with DT in IoT systems.

(iii) Application of ISAC-DT enabled IoT systems

Figure 2 presents the common application of IoT systems. The IoT system is made up of BS and target. The target completes duties such as sensing environmental information. The BS, which has the ISAC capabilities, senses the environment and the target. The sensor information is then exchanged with the target to ensure its safety. The DT contains sensing information, allowing for efficient environmental sensing, communication, networking and target management.

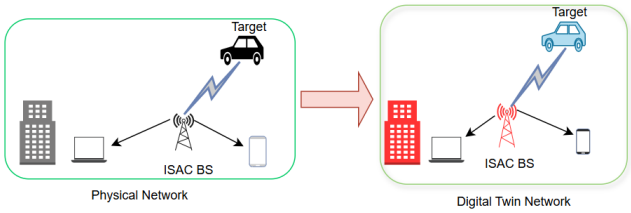


Figure 2. Application of ISAC-DT enabled IoT systems.

2.2.2. Performance Metrics

The evaluation of communication performance in ISAC systems involves the uses of various performance metrics, including spectral efficiency (SE), outage probability, and energy efficiency, which provide insights into the effectiveness and reliability of signal transmission. Conversely, the evaluation of sensing performance involves metrics, such as the probability of detection (PD), spectral efficiency, signal-to-interference-plus-noise ratio (SINR), Cramer-Rao Bound (CRB), and energy efficiency. Figure 3 illustrates the performance metrics associated with ISAC system.

(i) Communication system performance metric

- Spectral Efficiency: Determines the maximum information transmission rate [22,30].

$$R = B\log_2(1 + SNR) \quad (5)$$

B represent the signal bandwidth, and signal-to-noise ratio (SNR) denotes the power of the received communication signal.

- Outage probability: The probability that a communication system fails to attain a designated data transmission rate under specified channel circumstances is expressed as [9]

$$P_{out} = P(R < R_{th}) \tag{6}$$

R denotes the data transmission rate, while R_{th} signifies the lowest data transmission threshold.

- Energy Efficiency: This metric represents the ratio of the achievable rate to the overall power consumption, expressed as [16]

$$EE = \frac{R}{P_{total}} \tag{7}$$

where P_{total} denotes the total power consumption.

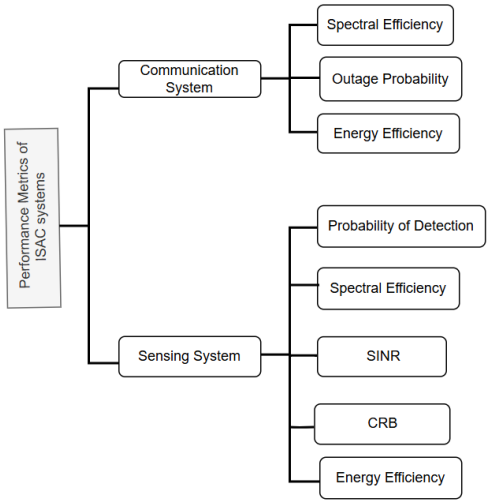


Figure 3. Performance metrics of ISAC system.

(ii) Sensing system performance metric

- Probability of detection: The likelihood of accurately recognizing or detecting the target under particular conditions is expressed as [20]

$$P_D = P(\Gamma > \mu | H_1) \tag{8}$$

Γ represents the likelihood ratio under hypothesis of a target's presence (H_1) vs the absence of a target, whereas μ denotes the detection threshold determined by the specific false alarm probability.

- Cramer Rao Bound: The efficacy of the target estimation for the unbiased estimator is presented as [23]

$$\mathbf{CRB} = \frac{\sigma_s^2}{2|\epsilon|^2||\dot{\mathbf{a}}||^2(||\mathbf{a}^H C||^2 + ||\mathbf{a}^H S||^2)} \tag{9}$$

$\dot{\mathbf{a}}$ is the derivative of $\mathbf{a}$, whereas C and S signify the communication and sensing beamforming, respectively. σ_s represents the noise power at the transmitter, while ϵ denotes the reflection coefficient of the target.

- SINR: The quality of the received sensing signal, in relation to interference and noise, is commonly quantified in decibels (dB) and described as [25]

$$\mathbf{SINR}\ (\mathrm{dB}) = 10 \log_{10}\left(\frac{Ps}{Pn}\right) \tag{10}$$

where *Ps* represents the power of the sensing signal and *Pn* is the power of noise and interference.

2.3. Digital Twin: Concepts and Structure

A digital twin is a model or up-to-date picture of a real-world item that is in use. It shows the present state of operations and may contain pertinent historical data [31,32]. The fundamental goal of digital twin is to evaluate the operation's current status and, even more crucially, to predict how it will act in the future, therefore, improve control or make it better. For example, smart cities, power plant, IoT devices can have a digital twin that represent a single part, a group of systems that are concerned. Physics-based approaches and statistical models can both be used in digital twin models.

2.3.1. Structure of Digital Twin Systems

A standard DT concept model comprises three components: a physical entity (PE) in real space, a digital replica (DR) in virtual space, and the intra-twin communications that link the PE to DR through current data and information [33–35]. The functions and characteristics of each part are described as follows [36,37].

- PE: In the physical domain of a DT system, many infrastructures, including sensors, cameras, are tasked with gathering data on the present physical data measurements of PE. An example is illustrated in Figure 4, where the IoT system serves as a tangible PE framework that includes various sensors, such as cameras strategically positioned to provide 360° view of the external environment, along with radar and LiDAR systems located on all four sides of the vehicles, which enable real time detection of surrounding objects and measurement of their distance and velocity.
- Virtual Representation of Digital Representation (DR): Within the virtual environment of the DT system, a DR functions as a software application that holds a real-time model of its PE. It obtains the real-time data gathered by PE through a synchronized private link, which may be either wired or wireless [38,39]. Upon processing the data, the DR is capable of not only visualizing and representing the current status of its PE but also calculating anticipatory operations to assist its PE in making decisions in the real world [36,37].
- Intra-twin communications: The relationship between PE and DR, namely the intra-twin communication, comprises two elements: the transmission of raw data and the processed information. The raw data refers to the information gathered by various sensors transmitted from the PE to the DR, while the processed information denotes the analytical results relayed from the DR to the PE. The connection between PE and its DR must be confidential and secure to facilitate the exchange of information and knowledge (or AI models) within the twin. The three components, namely PE, DR, and intra-twin communication, provide a comprehensive inner loop of the DT system that facilitates successful simulation and feedbacks mechanisms of the twin [36,37]. To provide additional detail, a comprehensive functional structure of DT with ISAC within the framework of the IoT is presented in Figure 4.

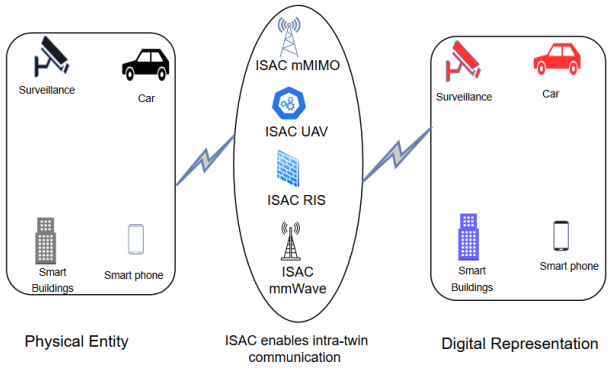


Figure 4. Architecture of ISAC enables intra-twin communication in a DT-driven IoT systems.

2.3.2. Architecture of Digital Twin Used in IoT Systems

The architecture of a digital twin, which is based on logical twin objects, is illustrated in Figure 5 [4,36]. Three layers can be distinguished in the architecture: The services layer which is often known as the cloud; the twin layer; and the physical device interface layer. In the services layer, you can locate application interfaces. Users can make service requests through a wireless system that utilises digital twins. Semantic reasoning frameworks translate the user's request and send the response to the twin layer. The twin layer comprises logical twin objects that utilise a digital representation of the physical object or phenomenon. A DT may use several models for physical things, including mathematical models, 3D models, and data-driven models [36,37]. The final layer is the physical interaction layer of the devices, encompassing all physical components, including end devices, edge cloud/servers, and network devices, among others. Efficient interfaces are essential between several layers in twin-based wireless communication. The interfaces include twin-to-physical object, twin-to-twin, and twin-to-service interfaces. To enhance the system total rate, the authors of [37] suggested a DT-based scheme for simultaneously designing the access point and user association, power control, and RIS beamforming. The study in [38] introduces a DT-enabled wireless system intended to energise several interconnected sensor nodes within an extensive IoT network architecture. The research in [39] presents an alternate optimisation approach for sensing and communication to mitigate the system overhead of digital network twin within a digital twin edge network. The research in [40] introduced a DT-based system to examine resource allocation for vehicular communications supported by UAV and reconfigurable intelligent surface. The authors of [41] presented a cyber-twin-driven integrated communication system for satellite-terrestrial vehicles to improve the physical layer security of satellite-terrestrial networks. To tackle the issue of offloading modifications in mobile edge computing (MEC) settings within the scope of the internet of vehicle (IoV), the authors [34] proposed a digital twin strategy to reduce energy consumption and latency in 6G IoV architecture. In a cell-free ISAC multiple-input multiple output (MIMO) system, a DT-enabled channel estimation approach is suggested for target sensing channel estimation in [36]. In order to facilitate RIS-assisted wireless communications, a DT of the physical environment was presented in [42] to approximatively model channels utilising an electromagnetic 3D model and ray tracing. The authors of [43] presented a DT-based THz signal guiding and control system to model, predict, and control the proprieties of signal propagation within buildings. In addition, to ease the burden of data collection in practical systems, the authors of [44] investigate DT's potential application in comprehensive sensing based on deep learning for hybrid precoding.

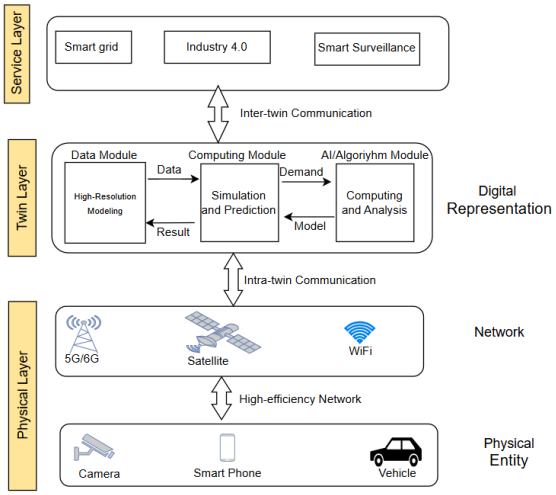


Figure 5. Architecture of a DT-enabled IoT system.

2.3.3. Advantages of Digital Twin

DT technology is regarded as critical in many fields, such as ISAC-based IoT systems, primarily because it provides numerous benefits, including reducing errors, uncertainties, inefficiencies, and costs in any system. Some of the benefits stated for DT include:

- Reliability: The digital twin technology utilises a store-and-forward (SF) networking technique to tackle challenges, including intermittent connectivity, long or variable delays, asymmetric data rates, and high error rates [45]. SF is a networking approach that involves the temporary storage of data packets at intermediate points within the network prior to their forwarding to the intended destination. This technique allows digital twin technology to handle problems caused by unstable network conditions, making sure data gets delivered successfully even in tough situations and improving the reliability of IoT networks.
- Optimising solutions: By storing and sending signals, the digital twin handles different delays and uneven data speeds, making communication more efficient in a complicated IoT system [31].
- Resiliency: A digital twin can improve the robustness of an IoT system, increasing its effectiveness in dealing with environmental uncertainties and operation failures [46].
- Synchronisation and adaptation: Digital twins facilitate swift recovery and adjustment to evolving conditions by preserving synchronised digital replicas of network elements and operations [5,45].
- Maintenance: Digital twins serve as crucial references for restoring operations and reconfiguring network resources during network failure or disturbances, effectively minimising outages and service disruptions.
- Predicting Solution/System Planning: By using past data and predictive analytics, digital twins can foresee possible failures and take steps in advance, like changing routes and adjusting resources, to keep services running smoothly and reliably.

2.4. Potential Benetifs of Combining DT and ISAC with IoT Systems

As described in Section 2.1, IoT systems confront various difficulties, including ineffective interconnectivity, spectrum congestion, energy inefficiency, excessive communication latency, and security. DT and ISAC provide practical solutions to these challenges by boosting connection, maximizing spectrum utilization and energy efficiency, and lowering latency through the application of essential communication technologies to facilitate the physical-to-virtual connection in DT and ISAC for IoT systems [24,37]. Table 1 illustrates the key characteristics of DT and ISAC, as well as their potential impact on IoT systems [36,38,39].

Table 1. Primary characteristics of DT and ISAC potential impact on IoT systems.

Characteristics of DT	Characteristics of ISAC	Joint Supplement on IoT Systems
Reduce latency	Enhance sensing accuracy rate	Facilitate real-time monitoring of sensing data and establish a reliable and secure sensing and communication channel collaboratively.
Energy efficiency	Spectral efficiency	Reduces interference and spectrum congestion, enhances system performances, and improves cos-effectiveness concurrently.

3. Related Works

This section analyzes and compares several review papers related to our research area to evaluate the implementation of DT and ISAC for IoT systems, while also identifying research gaps, shortcomings, and strengths within these studies. The review in [2] outlines the principles of DT for wireless systems, including taxonomy and outstanding research difficulties within the wireless communication framework. However, the review did not consider the integrated sensing and communication technologies for intra-twin communication. The survey in [3] offers a review of the most recent definitions of DT, an examination of the key attributes that DT ought to have, and a look at the fields where DT applications are being created. Nonetheless, the implementation of ISAC with DT in IoT systems was not addressed. To fully understand what DT technology is supposed to do, the survey in [4] gave an overview of DT technology and how it is used in other new areas, like augmented and virtual reality (AR/VR). Nevertheless, the survey did not consider the communication technologies facilitating physical-to-digital communications. The review in [6] presents a thorough investigation of how AI and DTN can be integrated into 6G communication systems. This study only examines the integration of AI and DT in 6G. The authors of [7] examine the digital twin network, analyze the evolution of digital twins, and classify significant technologies, functional frameworks, and applications of the Internet of Digital Twin (IoDT) communication. This study did not address application of ISAC technology in the IoDT. The authors of [47] provide a comprehensive review of digital twin networks (DTN), highlighting significant use cases and their service requirements, outlining a standard architecture, and examining fundamental design considerations for 6G wireless networks. In [48], a survey was shared that summarised how DT can be used in sixth-generation (6G) communication systems by introducing a framework based on DT for 6G internet of everything (IoE) applications. Nonetheless, this study did not examine the application of ISAC technology in the development of 6G. The authors of [49] examine the application of DT technology in 6G communication systems, offering an analysis and overview of its utilization while identifying research obstacles and necessary future actions for the complete integration of DT technology in 6G communication systems. The application of ISAC and DT in IoT systems, however, was not investigated. In [50], the authors present a vision and elucidate various methodologies for developing and employing a real-time digital twin for future wireless communication and sensing systems, along with prospective research trajectories. In [51], the survey lays out the fundamental concepts of DT communication and identifies the areas that still require further investigation. The survey referenced [52] provides an overview of IoDT, covering topics such as system architecture, enabling technologies, and security/privacy concerns. The survey did not include the communication technology facilitating intra-twin communication. The review in [53] presents the concept of a 6G space-air-ground integrated network (SAGIN) enabled by cybertwins, aiming for deep integration and pervasive intelligence within the 6G SAGIN framework. The authors of [54] introduced a DT structure based on ISAC and the enabling technologies for networks of intelligent machines.

This survey differentiates itself from the previously reviewed articles on DT by focusing on a comprehensive examination of the fundamental characteristics and essential enabling technologies for DT and ISAC applications in IoT systems that promote communication between the physical entity and its digital counterpart. This work encompasses an analysis of the main facilitators and the problems that emerge. Table 2 compares the contributions of our survey with those of earlier survey research in the field of DT and ISAC for IoT systems, focusing on the year of publication, communication technology facilitating physical-to-virtual connections, and limitations.

Table 2. Comparing our work to pertinent surveys.

Years	Refs	RIS	UAV	THz/mmWave	mMIMO	Drawbacks
2019	[3]	No	No	No	No	The application of DT in the communication sector was not addressed.
2021	[4]	No	No	No	No	Concentrate on the key technologies for digital twin networks. Furthermore, the subject of major communication facilitators for digital twins has not been examined.
2021	[51]	No	Yes	No	No	Only one communications technology for the physical-to-virtual connections was investigated.
2022	[2]	Yes	No	Yes	No	There was no discussion of communication methods for DT of wireless communication.
2022	[48]	No	No	Yes	No	The communication technology used between the physical layer and its digital twin layer in the DT architecture was not taken into account.
2022	[49]	Yes	No	Yes	No	The implementation of DT in IoT-enabled ISAC application was not considered.
2023	[50]	No	No	Yes	No	Concentrate primarily on the various approaches for building a DT for 6G and beyond, without addressing the use of DT in IoT systems.
2024	[6]	Yes	Yes	Yes	No	ISAC use was not considered when integrating artificial intelligence and digital twin networks.
2025	[7]	Yes	No	Yes	No	The ISAC techniques for the intra-twin communication in the internet of digital twin scenario were not examined.
Now	**Ours**	**Yes**	**Yes**	**Yes**	**Yes**	**Extensive research has not yet been conducted on novel concepts regarding the application of DT and ISAC in IoT systems.**

4. Survey Methodology

By employing the research goal formulation outlined in [55], one can guarantee transparency and a selection process that is replicable. The methodology consisted of five distinct steps: (1) articulating precise research questions, (2) devising a thorough search strategy, (3) defining explicit inclusion and exclusion criteria, (4) executing a meticulous quality assessment, and (5) subsequently performing a data extraction. Figure 6 illustrates the methodology for the systematic literature review in this study.

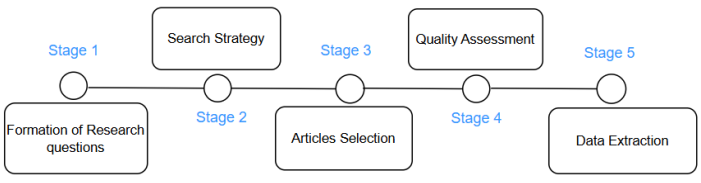


Figure 6. Steps of the Systematic literature review.

4.1. Search Strategy

An extensive search was conducted across multiple academic databases to ensure a comprehensive and systematic retrieval of relevant papers. The search strategy involved the selection of pertinent academic databases, followed by the formulation of search queries utilizing keywords and Boolean operators. To ensure transparency and replicability, only papers that were available online were examined. Subsequently, suitable search filters were applied to refine the results, and finally, all retrieved studies were systematically documented. This methodological approach ensures that the research is focused on investigating DT and ISAC applications in IoT systems, addressing open issues, and emphasizing essential communication technologies for their implementation.

4.1.1. Digital Databases

To ensure we found a lot of relevant literature, we chose five well-known databases that are frequently used in wireless communications, IoT technology, and DT research. This study looked at peer-reviewed journal articles, conference proceedings, and workshop papers that were published between 2019 and June 2025. We chose the database based on the fact that they have a wide range of high-quality, peer-reviewed articles such as systematic literature reviews. We sourced data from Web of Science or Scopus using the following databases: (1) IEEE Xplore, (2) ScienceDirect, (3) ACM, (4) SpringerLink, and (5) Google Scholar. However, the indexing method used Web of Science or Scopus for our study frameworks. We used the search query in Figure 7 and got 2689 articles back.

1. "Digital Twin" AND "IoT" OR "Internet of Things" in "IoT-aided Integrated Sensing and Communication"

2. "Digital Twin" AND "IoT" OR "Internet of Things" AND "ISAC" OR "Integrated Sensing and Communication" in "IoT-aided ISAC"

Figure 7. Keywords and Search strings for Systematic Review.

4.1.2. Search Query Formulation

To improve the efficiency of the search process, search queries were developed with the help of Boolean operators (AND, OR). The query illustrated in Figure 7 is utilised to interrogate the digital literature indexer. The objective is to encompass diverse types of digital twins and ISAC applications within the context of the IoT scheme.

4.2. Method of Articles Selection

The article selection method comprises five stages, as illustrated in Figure 8. This document details the number of papers that were obtained, screened and eventually incorporated into the final synthesis. A total of 2689 papers is selected initially. Consequently, a total of 1084 results were acquired IEEE, 486 from Springer, 786 from ScienceDirect, 93 from ACM, and 240 from Google Scholar.

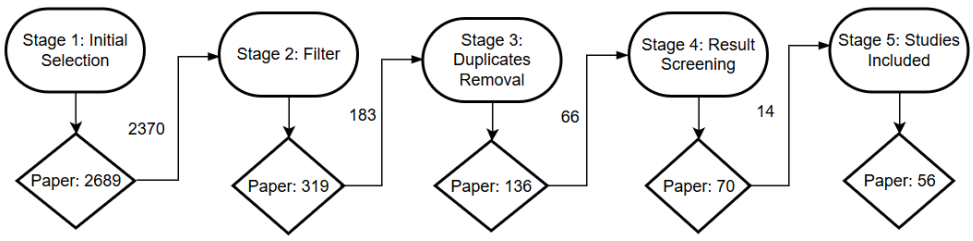


Figure 8. Method of article choosing process.

4.2.1. Inclusion and Exclusion Criteria

To achieve a rigorous study selection, we used inclusion and exclusion criteria, which allowed us to filter out and locate high-quality, peer-reviewed publications directly relevant to our research objectives and research questions; specifically, the types of studies that were excluded are books, magazines, and dissertations as shown in Table 3, because they were not peer-reviewed studies. Table 3 presents a complete review of the criteria. As a consequence, 2370 publications were discarded, leaving 319 articles for abstract screening.

Table 3. Inclusion and Exclusion Criteria for the Article Selection.

Inclusion Criteria	Exclusion Criteria
Journal articles, Conference papers, Workshops papers.	Book, magazines, dissertations.
The articles must be written in English.	Non-English articles.
Articles published between 2019 and June 2025.	Duplicates
The article is accessible with full text available.	Full text is inaccessible.

4.2.2. Abstract Screening

The 319 papers that passed the initial selection step were reviewed based on the relevance of their abstracts to the objectives of this study. At this point, each abstract was reviewed to determine how well it met the research questions, with focus on studies using DT and ISAC for IoT framework.

Following the review of abstracts, 183 papers were excluded, resulting 136 studies advancing to the full-text screening stage, where assessment were performed to verify compliance with inclusion criteria.

4.2.3. Full-Text Screening

At this step, each of the 136 papers was systematically evaluated to ascertain whether it provided significant new insights regarding our study approach. At this stage, 70 papers were kept, while 66 were rejected for the following reasons:

- The research did not specifically examine DT and ISAC for IoT systems.
- The papers did not address ISAC for IoT networks.
- The study did not address DT using the ISAC-enabled IoT systems.

4.3. Quality Assessment

A quality assessment was performed to ascertain that the chosen studies adhere to the necessary methodology, accuracy, and relevance, in accordance with the guidelines from [56] for systematic reviews. Every study experienced a rigorous evaluation based on established quality assessment criteria, thereby guaranteeing that only pertinent research was incorporated into the final analysis.

- Does the study design adequately address at least one of the research questions posed?
- Does the research examine ISAC within the context of IoT systems in the study?
- Is there an exploration of DT with the context of IoT systems in the study?
- Is the research objective and methodology articulated with clarity in the paper?

Following the quality assessment, 14 papers were eliminated, and after the application of all inclusion, exclusion, and quality assessment criteria, a final collection of 56 studies was chosen for review. After selecting 56 pertinent studies for our systematic literature review, we conducted a thorough data extraction and synthesis. Each study was examined and classified depending on its emphasis area, indicating whether the IoT systems were developed using digital twin scheme or ISAC techniques. We additionally categorised the articles based on whether they discussed DT- or ISAC-based IoT systems. The process

sought to extract all pertinent information to address the research question presented in Section "The Importance of Integrating DT and ISAC in the IoT Development". Figure 9 shows the classification of IoT system based publications.

Therefore, Figure 9 indicates that this study introduces a novel taxonomy that categorizes essential communication technologies facilitating physical-to-virtual communication for DT and ISAC for IoT systems, in contrast to the classifications found in earlier surveys.

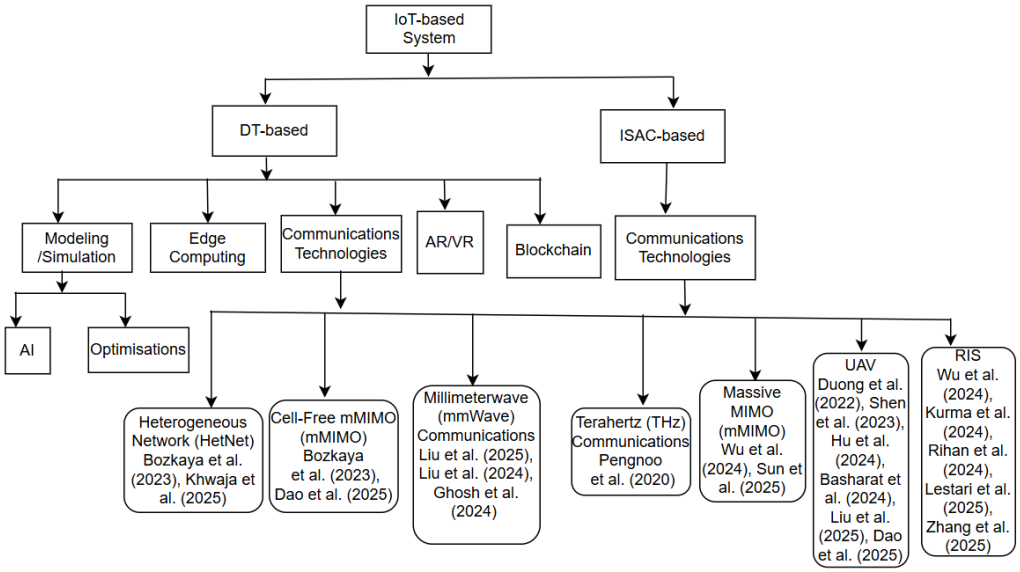


Figure 9. Taxonomy of IoT system based literature [43,57–85].

Comparative Analysis of Communication Technologies Used in DT and ISAC for IoT Systems

To further investigate the importance of communication technologies in the classification of IoT systems based on publication, we conduct an analysis of communication technologies utilized in DT and ISAC for IoT systems. Table 4 compares communication methods based on characteristics such as SE, EE, PD, latency, robustness, and range. Table 4 demonstrates that RIS technology is more EE and less costly to operate than UAV, mMIMO, mmWave, and THz technologies, while also exhibiting comparable latency to UAV, mMIMO, mmWave, and THz technologies. This distinction originates from the nature of RIS technology, which is a low-cost means of information transmission and a viable strategy for improving the integration of DT and ISAC in IoT systems.

Table 4. Comparison among communication technologies.

Reference	Technologies	Latency	SE	EE	PD	Robustness	Computational Cost	Range
[86]	RIS	Low	High	High	High	High	Low	High
[57]	UAV	Low	High	Low	High	High	High	Very high
[87]	mMIMO	Low	Very high	Low	High	High	High	Very high
[88]	mmWave	Low	High	High	High	Low	High	Low
[43]	THz	Low	Extremely high	Low	High	Low	High	Low

5. Results

This section discusses the results obtained for each research question mentioned in Section "The Importance of Integrating DT and ISAC in the IoT Development".

5.1. Evolution in Publication Types

This subsection intends to address our research question (1): How have publication types connected to DT and ISAC for IoT systems evolved over years? To answer this

question, we look at the current research inquiry, which includes the years 2019–June 2025 and focuses on 56 articles on DT and ISAC for IoT systems.

Figure 10 shows a distribution of articles organized by publication type (journal, conference, workshop) from 2019 to June 2025. The graph reveals minimal growth during the initial year (2019–2021), although the general publishing trend ascends throughout the year, resulting in a peak of 16 articles in June 2025.

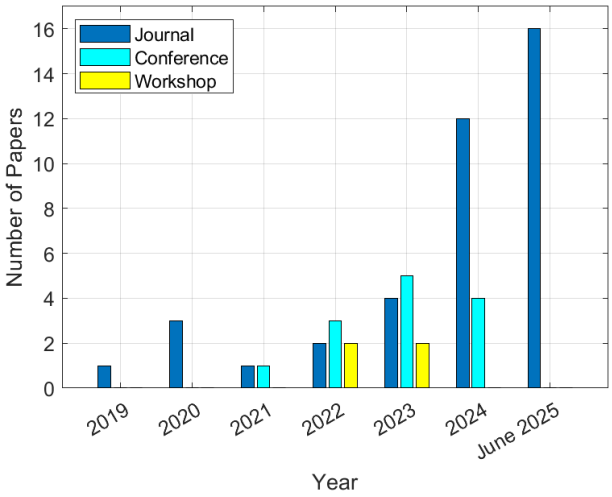


Figure 10. Distribution of articles by publication type.

5.2. Distribution of Articles

The second (2) and third (3) research questions: What types of publications have researchers used to investigate DT and ISAC for IoT systems? Which digital databases contain the most articles on this topic? We address these questions by analyzing the distribution of 56 papers across four digital databases and various publication types.

Figure 11 illustrates the distribution of research papers, categorized by publication type, within the DT and ISAC for IoT systems.

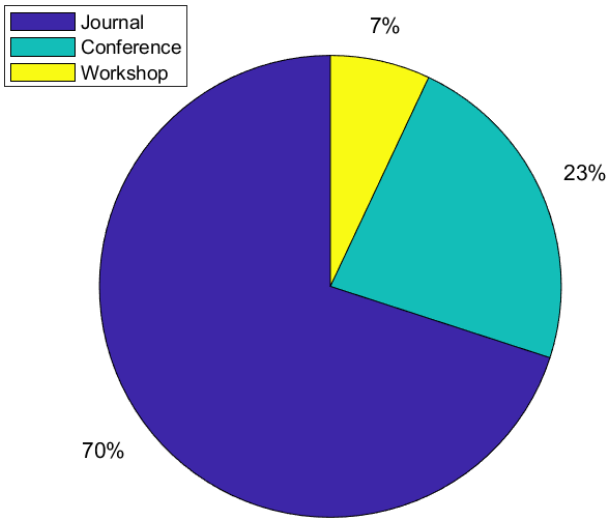


Figure 11. The distribution of research papers by publication type.

Figure 12 demonstrates the distribution of 56 papers across various publishing databases. IEEEXplore comprises the largest share of publications (66%), with Springer at 18% and ScienceDirect at 11%, and ACM holding a total of 5%.

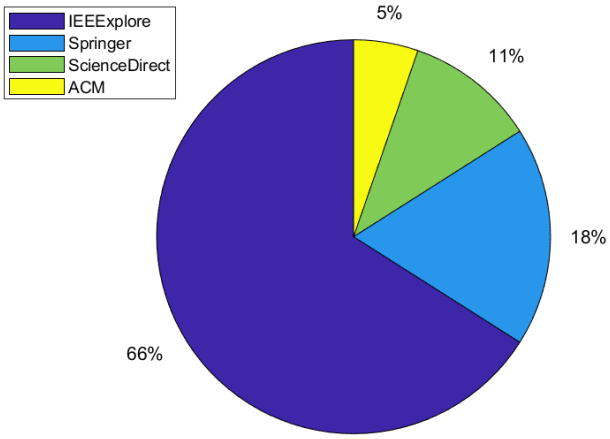


Figure 12. Distribution of articles in various databases.

5.3. Key Enabling Technologies for Physical-to-Virtual Connections in DT and ISAC for IoT Systems

This section aims to answer our fourth research question (4): What are the essential communication technologies that enable physical-to-virtual connections in DT and ISAC for IoT systems? To address this question, we investigate how different essential communication technologies have been studied in the DT and ISAC for IoT frameworks. Specifically, we examine the number of papers focusing on each essential communication technology and whether they target ISAC-based or DT-based schemes.

Figure 13 shows an overview of the distribution of communication technologies in DT and ISAC for IoT systems research. The subplot (a) depicts the distribution of communication technologies in an ISAC-based system, whereas the subplot (b) highlights studies concentrating on a DT-based system. Our findings show that RIS, mMIMO, CF-mMIMO, UAV, THz, mmWave, and HetNet technologies are the most extensively researched in DT and ISAC for IoT systems. However, there is a significant difference in their distribution between ISAC-based and DT-based systems. The most often communication technologies examined in the ISAC-based systems were RIS, mMIMO, and UAV, as demonstrated by six and four studies, respectively. THz and mmWave technologies were investigated in one and three articles, respectively, but CF-mMIMO and HetNet were absent from DT and ISAC for IoT systems.

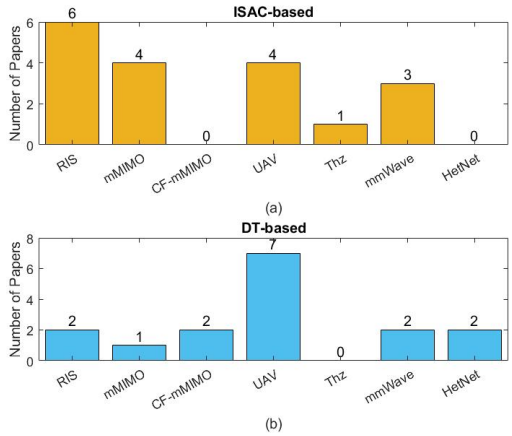


Figure 13. Communication technologies study distribution in an ISAC-based scheme (**a**) and a DT-based scheme (**b**).

DT-based systems lacked representation in communication technologies. The 16 studies on DT and ISAC for IoT systems focused mostly on UAV (7 papers), followed by RIS,

CF-mMIMO, mmWave, and HetNet technologies (2 papers each), while only one study addressed mMIMO technology. THz technology is missing from DT and ISAC for IoT systems research, indicating a significant gap in DT and ISAC for IoT development of underrepresented communication technologies.

5.3.1. RIS Technology

Several studies emphasize the application of RIS in ISAC-based and DT-based systems, especially in the context of DT and ISAC for IoT systems. The authors in [89] have explored the simultaneously transmitting and reflecting RIS aided ISAC within the an IoT system. An alternating optimization scheme was introduced to develop the active and passive beamforming aimed at maximizing the minimum SINR of the network. The authors of [86] further explore the application of RIS, specifically focusing on a sub-connected RIS-assisted mobile edge computing system for ultra-reliable and low-latency communication (MEC-URLLC) within the a DT-enabled context. They demonstrate that, although sub-connected RIS enhances latency and task offloading performance, fully active RIS exhibits inferior performance within DT-enabled URLLC system. The authors of [90] presented a joint optimization of transmit, receive, and reflection beamforming variables for a double active RIS-assisted ISAC system, with the objective of maximizing the minimum SINR across multiple targets over a sequence of time slots. Their findings indicated that active RIS technology enhances network performance compared to its passive version, while the generative adversarial network-based twin delayed deep deterministic policy gradient (GAN-TD3) method surpasses the traditional TD3 method, although it requires increased computing complexity. These studies highlight the benefits of designing RIS beamforming to enable DT and ISAC for IoT systems which effectively utilize the advantages of RIS technology in conjunction with ISAC-based or DT-based systems.

5.3.2. UAV Technology

UAV technology constitutes another important communication technique for constructing DT and ISAC for IoT systems. The authors in [91] presented a digital twin-assisted task offloading scheme within the UAV-MEC network. To increase the number of related IoT devices and decrease latency, a branch-and-bound method was presented for designing the placement of UAVs and IoT device associations. The authors of [92] developed a multi-step learning technique based on Dueling Double Deep Q-Learning (DDQN) to optimize UAV trajectory, aiming to enhance both UAV flight duration and overall radar estimation rate for UAV-supported IoT system. These studies underscore the significance of UAV placement and trajectory design, along with the necessity to improve communication quality between the physical entity and its digital representation in DT and ISAC for IoT systems.

5.3.3. Millimeter Wave Communication (mmWave) Technology

mmWave communication technology remains a critical concern in DT and ISAC for IoT system, particularly for ISAC-based systems. The research presented in [88] details a passive relay RIS-assisted mmWave ISAC system designed for multiple input and single output (MU-MISO) communications and multi-target sensing within IoT systems, focusing on improving the spectral efficiency of the system. The authors in [93] presented a mmWave utilizing a rate splitting multiple access (RSMA)-based ISAC technique to enhance the spectrum efficiency of RSMA-based ISAC system. According to these studies mmWave technologies can improve the performance of ISAC systems.

5.3.4. mMIMO Technology

mMIMO technology presents another significant communication technology for developing physical-to-virtual connections in DT and ISAC for IoT systems. The authors [94]

proposed an optimal transmit covariance and on-off control strategy in semi-closed form to enhance energy efficiency at a mMIMO transceiver the ISAC system. In [87], the authors present a joint alternating optimization and fractional programming method for formulating an ISAC joint rate within a non-orthogonal multiple access (NOMA)-based ISAC system in a near-field extreme MIMO environment.

We found that a significant 71% of the 56 studies we reviewed on DT and ISAC for IoT system focused on ISAC-based systems, primarily emphasizing UAV, RIS, and mMIMO technologies. On the other hand, merely 29% (16 papers) of the studies investigated communication technologies within DT-based systems, with a predominant focus on UAV technology (43.7%). Only 12.5% of the DT-based systems examined focused on RIS, CF-mMIMO, mmWave, HetNet, and 6.3% mMIMO technologies, with no investigations into THz communication technology. The findings highlight the necessity for effective sensing and communication strategies in DT and ISAC for IoT systems, aimed at enhancing communication and sensing performance between the physical entities and their digital representation of the DT and ISAC for IoT systems.

5.3.5. Summary for Research Question (4)

Figure 13 illustrates the increased prevalence of UAV technology in DT-based systems, highlighting its capacity to enhance the reliability of channel connectivity, improve channel capability, and boost data transfer rates for both the physical entity and its digital counterpart. On the other hand, ISAC-based systems prioritize technologies like RIS and mMIMO. This difference arises because ISAC-based systems utilize RIS and mMIMO technologies to provide significantly enhanced sensing resolution and accuracy as well as mitigate interference, enhance signal power, and enhance system security. Therefore, integrating DT alongside ISAC can further enhance IoT systems. This integration will help to enhance connectivity, security, and real-time sensing data transfer rates for communication between the physical entity and its digital counterpart.

5.4. Analysis of DT and ISAC Performance Metrics in IoT Systems

To further their developing capabilities and implications for IoT, we provide an analysis of important performance metrics across DT and ISAC. Performance metrics like SE, EE, PD, latency, and age of information (AoI) are visualized in bar chart (see Figure 14), which offers a comprehensive picture of the most performance indicators in DT and ISAC for IoT research.

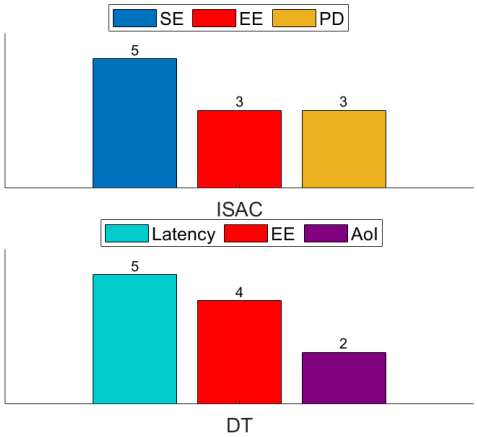


Figure 14. DT and ISAC performance metrics.

5.4.1. EE

EE, quantified in bits per joule, indicates the system's capacity to transmit data with little energy expenditure [80]. A greater number of studies concentrate on the EE issue in DT and ISAC for internet of things research [94,95]. This performance indicator facilitates more sustainable and economical DT and ISAC for IoT system operation.

5.4.2. Latency

For real-time digital twin applications, latency, the delay in data transmission, is essential. There are roughly five (5) studies that discuss latency in DT. For DT applications that require real-time responses, such as DT and ISAC applications in the IoT systems, this focus is crucial [96].

5.4.3. Age of Information

The AoI is an important measure in digital twin applications for the IoT. It evaluates the timeliness of the data a system possesses about the real-world object under control [97,98]. Only two publications look at the AoI metric in DT for IoT systems. This feature highlights a limitation in improving AoI performance in DT and ISAC for IoT systems research.

5.4.4. Spectral Efficiency

The SE represents the amount of information that can be reliably transferred per complex-valued sample and is crucial in ISAC applications for IoT [87]. A growing number of studies focus on enhancing the SE metric in this context. Improving this metric is vital for supporting more data-intensive applications, such as high-quality video streaming and real-time data transmission, which are required for the operation of the internet of things. Therefore, improving SE in the DT and ISAC application for IoT systems is crucial.

5.4.5. Probability of Detection

PD is a critical performance metric for ISAC applications in IoT systems, as it assesses the sensing capabilities of real-time data generated by the ISAC system. Three research papers investigate the concept of PD within the context of ISAC applications for IoT systems [99]. In scenarios where accurate real-time sensing is essential, such as digital twins and integrated sensing and communication for IoT systems, it is vital to consider the PD metric.

5.4.6. Implications of DT and ISAC Application in IoT Development

The enhancement of these performance metrics has a direct impact on the suitability of DT and ISAC applications in the development of IoT. For example, the high spectral efficiency and elevated probability of detection provided by ISAC are crucial for IoT systems that facilitate real-time object detection and enhanced device connectivity [92,99,100]. On the other hand, DT capabilities are advantageous for IoT systems because they have low latency and high energy efficiency, which makes it easy for the physical entity and its digital counterpart to communicate with each other [95,101].

ISAC-based systems utilize IoT transceivers for sensing and communication, potentially leading to energy inefficiency. The implementation of DT enhances the latency and energy efficiency of IoT systems through a synchronized and coherent operation between the physical entity and its digital counterparts. DT and ISAC function as the fundamental components for IoT systems, facilitating low latency, widespread, and secure, as well as energy-efficient, data sensing and communication that are crucial for IoT systems. The integration of DT with ISAC technologies is essential for IoT systems, facilitating seamless connectivity for millions of IoT devices while achieving energy efficiency and security standards. The combination of DT and ISAC for IoT systems reduces latency and energy

use while improving spectral efficiency in the communication between physical and digital objects. Table 5 presents a comparative analysis of ISAC and DT technologies.

Table 5. Comparison of ISAC and DT technologies.

Technologies	Advantages	Disadvantages
ISAC	• Enable seamless communication and accurate sensing to establish connectivity between physical entities and their digital representations. • Enhance the security of IoT system while improving spectrum utilization efficiency.	• Sensing and communication functions necessitate a high energy consumption
DT	• Enhance energy efficiency and reduce latency by creating connected physical and digital objects. • Increase cost-effectiveness and enable predictive maintenance.	• The communication channel used to connect the physical entity and its digital representation is both unsafe and unreliable. • the intra-twin connection suffers from poor spectral efficiency, which affects real-time sensing and communication.

6. Open Challenges and Recommendations

As a emerging technology, limited study has been undertaken on the physical-to-virtual communication technologies of and DT and ISAC for IoT systems. Numerous unresolved challenges in DT and ISAC for IoT systems require attention. This section examines and highlights various unresolved challenges in the advancement of DT and ISAC for IoT systems.

6.1. Interference Management

6.1.1. Integration of Various Frequency Bands

Due to the distinct applications associated with various frequency bands, thorough optimization is necessary to reduce interference. Low-frequency communication signals exhibit significant penetration and coverage capabilities; however, they are constrained by insufficient sensing precision and time-frequency resolution. With increasing frequency, shorter wavelengths and wider bandwidths result in improved sensing accuracy and higher communication data rates. Integrating the complementary advantages of mmWave, THz, and visible light technology in DT and ISAC for IoT systems can effectively mitigate interference issues in single-frequency DT and ISAC for IoT operation.

6.1.2. Novel Degree of Freedom

RIS and UAV provide a new dimension for interference management. RIS enables precise signal control, while UAVs enhance system capabilities by facilitating an agile LOS path. The integration of RIS enables the system to utilize controlling signal reflection and refraction. RIS can significantly decrease interference at a low cost while establishing virtual LOS connections, which are crutial for extensive sensing networks in environmental monitoring [42].

Conversely, UAV-assisted DT and ISAC for IoT systems offers considerable potential because of the mobility and adaptability of UAVs. Nonetheless, it faces several challenges. UAVs face limitations due to constraints in information processing capabilities, energy supply, and communication interactions. As a result, current solutions frequently fail to adequately address the sensing and communication needs, especially in dynamic and resource constrained deployment context [9]. A promising research direction involves the use of UAVs as a mobile platforms for efficient target tracking with reduced power consumption.

6.2. Connectivity

Connecting a physical entity to its digital representation system is another significant problem, particularly when a large number of IoT sensors must be connected simultaneously and a real-time control is required. Indeed, because sensors and, more broadly, IoT devices serve as source of data for artificial intelligence (AI) algorithms, missing IoT data from one or more devices due to connectivity failure may have an impact on data mining accuracy and, ultimately, system performance. Then, a high level of connectivity and a reliable solution to prevent power failures, coverage outages, and software mistakes must be devised.

6.3. Mobile Users Handover Management

Mobility represents a significant challenge that requires attention, particularly within the framework of DT and ISAC for IoT systems. Continuous and seamless connectivity must be ensured for IoT devices to prevent service interruptions when a device moves outside the coverage area of the associated BS. Consequently, it is essential to develop a highly accurate mobility prediction model, along with efficient solutions for both the model and the transfer of learning from one digital twin to another, prior to the physical entity's connection to the new DT.

6.4. Energy Efficiency

Although there have been numerous intriguing studies aimed at enhancing the energy efficiency of IoT networks, this issue continues to pose a major obstacle. There is significant potential for in-depth discussion regarding sustainable IoT networks. The proliferation of devices and heightened connectivity exacerbate the challenges faced DT and ISAC for IoT systems. An intriguing option is the use of renewable-powered green UAVs; mMIMO, RIS, and CF-mMIMO technologies could enhance energy efficiency and coverage for DT and ISAC for IoT systems. In summary, we believe that no single technology is likely to meet all the requirements at the same.

6.5. Security

The implementation of DT and ISAC for IoT system would encounter several security challenges owing to its intricate and interrelated characteristics. DT is engineered to emulate real-time physical item environments within virtual settings; consequently, DT must manage substantial volumes of sensitive data and enable essential interactions between physical and digital entities. This renders DTs and ISAC applications vulnerable to numerous threats and vulnerabilities that might destabilize the integrity, confidentiality, and availability of information [102]. A key security worry is the potential for data breaches. DT must securely store and process sensitive data pertaining to sensing and communication. If unauthorized entities access this essential information through a cyberattack, it may result in significant security concerns. Furthermore, recognizing access control is essential for DT applications in the IoT to avert unwanted access. Therefore, effective authentication procedures are essential to ensure that only authorized users can interact with the DT system. Artificial intelligence and metaheuristic methodologies present prospects for improved security in DT and ISAC for the IoT systems [103,104]. AI-driven secure DT and ISAC for IoT applications can assist in identifying anomalous activity that may signify a security breach in the system.

6.6. Privacy

Privacy is a significant concern in DT and ISAC for IoT systems, especially when handling personal information. Adhering to data protection regulations is vital to protect

users' privacy and prevent unauthorized access to personal data [6]. To tackle these privacy challenges, the decentralized, immutable, and transparent nature of blockchain technology can be utilized to enhance data integrity and privacy in the context of DT and ISAC for IoT systems [105]. Information related to DT and ISAC in the IoT ecosystem can be recorded in a blockchain block and linked through cryptographic hashes, which guarantees both immutability and transparency [106]. The distribution structure of blockchain minimizes the risk of single points of failure and bolsters resilience against cyberattacks. Implementing blockchain in the context of DT and ISAC for IoT systems strengthens data privacy and trust, thereby ensuring effective security and privacy controls.

7. Conclusions

This study investigated the communication methods employed between the physical entity and its digital representation for DT and ISAC for IoT systems. We specifically focused on RIS, UAV, mmWave, and mMIMO technologies. Initially, we delineated the fundamental concepts for implementing DT and ISAC for IoT systems. Moreover, we highlighted how our study differs from existing related survey studies. Furthermore, we presented our survey technique. Subsequently, we presented comprehensive responses to our four research questions: (1) How have publication types connected to DT and ISAC for IoT systems evolved over the years? (2) What types of publications have researchers used to investigate DT and ISAC for IoT systems? (3) Which digital databases contain the most articles on this topic? (4) What are the essential communication technologies that enable physical-to-virtual connections in DT and ISAC for IoT systems?

Ultimately, we have delineated significant challenges and prospective avenues for further research. Furthermore, the future directions within the field of DT and ISAC for IoT systems that require attention are as follows:

- Research combining generative AI with technologies such as blockchain to enhance security and privacy of DT and ISAC for IoT systems.
- Integrating a hybrid UAV-RIS technology with metaheuristic methods to reduce interference and improve channel quality between twins for DT and ISAC for IoT systems.
- Research on hardware architecture for energy-efficient DT and ISAC applications in IoT systems.

This article may function as a reference for research on facilitating communications for physical-to-virtual connections within DT and ISAC for IoT systems.

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